

# Alur Studi



# Kurikulum

| 1st Year                               |         |                                        |         |
|----------------------------------------|---------|----------------------------------------|---------|
| 1st Semester                           | Credits | 2nd Semester                           | Credits |
| Pancasila                              | 2       | Introduction to Digital Transformation | 2       |
| Cvics                                  | 2       | Linear Algebra                         | 2       |
| Calculus 1                             | 3       | Calculus 2                             | 3       |
| Physics 1                              | 4       | Physics 2                              | 3       |
| Chemistry                              | 3       | Laboratory Physics Experimental        | 1       |
| Engineering Drawing                    | 3       | Probability Theory                     | 3       |
| Introduction to Industrial Engineering | 2       | Engineering Mechanics                  | 2       |
|                                        |         | Engineering Materials                  | 2       |
|                                        |         |                                        |         |

| 2nd Year                |         |                                     |         |
|-------------------------|---------|-------------------------------------|---------|
| 3rd Semester            | Credits | 4th Semester                        | Credits |
| Statistics              | 4       | Mechatronics                        | 2       |
| Calculus 3              | 3       | Measurement & Design of Work System | 3       |
| Computer Programming    | 3       | Organizational Behaviour            | 2       |
| Operations Research 1   | 3       | Operations Research 2               | 3       |
| Manufacturing Process   | 3       | Economics & Engineering Economics   | 3       |
| Cost Analysis & Control | 2       | Data Analytics                      | 3       |
|                         |         | Production Planning & Control 1     | 2       |
|                         |         | Industrial Ecology                  | 2       |
|                         |         |                                     |         |

| 3rd Year                                    |         |                                         |         |
|---------------------------------------------|---------|-----------------------------------------|---------|
| 5th Semester                                | Credits | 6th Semester                            | Credits |
| Ergonomics                                  | 2       | Indonesian                              | 2       |
| Production Planning & Control 2             | 3       | Supply Chain System                     | 2       |
| Quality Control & Assurance                 | 3       | Occupational Safety & Health            | 2       |
| System Simulation                           | 3       | Product Design & Development            | 2       |
| System Modelling                            | 2       | Analysis & Design of Information System | 3       |
| Industrial Organization Design & Management | 3       | Business Planning                       | 3       |
| Facilities Design                           | 3       | Research Methodology                    | 2       |
|                                             |         | Integrated Laboratory Work              | 2       |
|                                             |         |                                         |         |

| 4th Year                                 |         |                   |         |
|------------------------------------------|---------|-------------------|---------|
| 7th Semester                             | Credits | 8th Semester      | Credits |
| English 1                                | 2       | English 2         | 2       |
| Entrepreneurship                         | 2       | Religions         | 2       |
| Elective Course 1                        | 3       | Final Project     | 6       |
| Elective Course 2                        | 3       | Elective Course 3 | 3       |
| Integrated System Design                 | 4       | Elective Course 4 | 3       |
| Implementation of Industrial Engineering | 2       |                   |         |
|                                          |         |                   |         |
|                                          |         |                   |         |
|                                          |         |                   |         |

| Elective Courses                     |         |                                                    |         |
|--------------------------------------|---------|----------------------------------------------------|---------|
| Courses                              | Credits | Courses                                            | Credits |
| Marketing Management                 | 3       | Service Quality Management                         | 3       |
| Quality Engineering & Management     | 3       | Financial Management                               | 3       |
| Enterprise Resources Planning System | 3       | Network Analysis                                   | 3       |
| Industrial Automation                | 3       | Advanced Production Planning & Control             | 3       |
| Human Resource Management            | 3       | Multivariate Analysis                              | 3       |
| Design of Experiment                 | 3       | Maintenance Management                             | 3       |
| Decision Analysis                    | 3       | Jig and Fixture & Matertial Handling System Design | 3       |
| Project Management                   | 3       | Corporate Strategy                                 | 3       |
| Meta-heuristic                       | 3       | System Dynamics                                    | 3       |